

VIBE2SHIP HACKATHON

Coding Ninjas × Google for Developers

NIVARAN

Hyperlocal Civic Issue Reporting Platform

निवारण — "resolution." Restoring trust between citizens and civic authorities.

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Problem Statement Selected

Community Hero — Hyperlocal Problem Solver

Indian communities deal with the same recurring problems — potholes, water leakages, damaged streetlights, and waste pile-ups. Reporting them is scattered across phone calls, WhatsApp groups, and disconnected municipal apps, so issues vanish with no visibility, no tracking, and almost no accountability.



Goal: let citizens report, validate, track, and resolve issues through community collaboration and intelligent automation — with transparency built in.



**Potholes &
Road Damage**



**Water
Leakages**



**Damaged
Streetlights**



**Waste
Management**

Why It Matters



78%

of citizens never get a follow-up after reporting an issue

LocalCircles, 2023



2,500+

lives lost to preventable hazards like open manholes (2017–22)

NCRB



₹6,000 Cr

spent yearly on pothole-related vehicle damage in India

ASSOCHAM



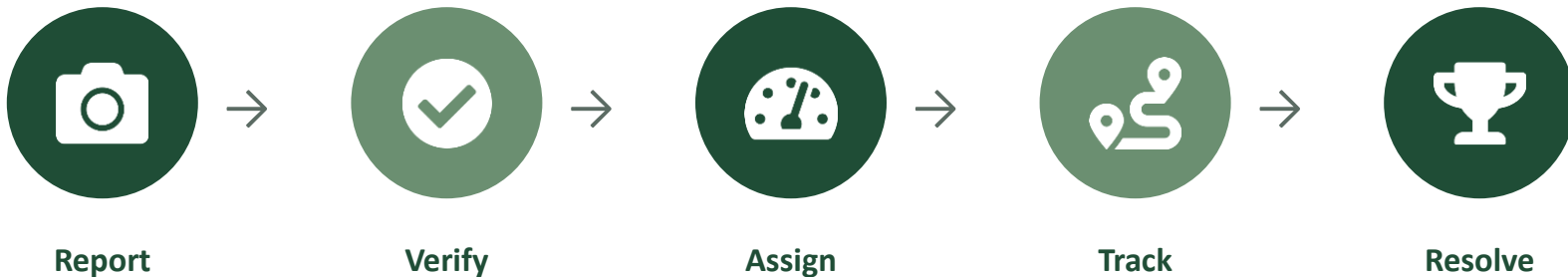
<12%

of Indians are fluent enough in English to use most civic apps

Census 2011

Solution Overview

Nivaran (Sanskrit: निवारण — "resolution") is a unified, AI-powered, multilingual platform that connects citizens directly to the civic resolution workflow.



Built for citizens, officers, and city administrators — each in their own language.

Citizen reports → community upvotes auto-verify → AI suggests the right officer → resolution tracked in real time

Key Features — For Citizens



AI Photo & Video Reporting

Snap a photo — Gemini auto-detects the category and writes the description.



Multilingual, Start to Finish

Full app interface in English, Hindi & Bengali — type or speak to report in your own language.



Emergency SOS Flagging

Life-threatening issues jump straight to the top of every officer's queue.



Gamification & Leaderboard

Badges and points turn civic reporting into a sustained community habit.

Key Features — For Officers & Community



AI Resolution Plans

Gemini drafts step-by-step fix plans and suggests the best-fit officer to assign.



Predictive Insights

Hotspot scoring and trend forecasting help administrators plan proactively.



Real-Time Status Tracking

A 5-stage timeline from Pending to Resolved, visible to citizens and officers alike.



Smart Duplicate Detection

Matching reports are merged automatically instead of piling up unreviewed.

Technologies Used



React 18 + Vite

Frontend



Express + Node.js

Backend



Firebase

Auth & Database



Tailwind CSS

Styling



Leaflet + OSM

Maps & Geocoding



Cloudinary

Media Storage



Recharts

Data Visualization



i18next

Localization



jsPDF

PDF Export

Google Technologies Utilized



Firestore Auth & Firestore

Secure sign-in and a real-time database powering reports, users, and notifications.



Gemini 2.5 Flash

Multimodal AI for image analysis, resolution-plan generation, and smart officer assignment.



Google AI Studio

Used to prototype and test Gemini prompts and structured outputs during development.



Web Speech API (Chrome)

Powers Hindi & English voice-to-text so citizens can report hands-free.

Projected Impact

40–60%

faster issue resolution

70%+

fewer duplicate reports

~90%

language coverage (vs ~12%)

5–10×

rise in community
participation

Estimates based on platform mechanisms — auto-triage, deduplication, multilingual support, gamification — not yet measured in live production.

Conclusion



Scalable

Firestore scales from hundreds to 100,000+ reports; Gemini and Cloudinary scale right alongside it.



Maintainable

Clear separation of concerns and role-gated routes keep a 10,000+ line codebase manageable.



Extensible

A modular architecture leaves room for real-time alerts, SMS, or smarter image classification.



User-Centric

Three-language support, voice input, and gamification meet citizens where they already are.

“Nivaran puts the power of civic resolution back in the hands of the community.”

Thank You